

Project Report : - Customer Segmentation Using Unsupervised Learning

K-Means Clustering on Mall Customer Dataset

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Github Repository Link : <https://github.com/jpSharma123-sudo/customer-segmentation-project>

Field	Detail
Project Type	Unsupervised Machine Learning
Algorithm	K-Means Clustering (K=5)
Dataset	Mall Customer Dataset — 200 records, 5 features
Tech Stack	Python · Pandas · Scikit-learn · Matplotlib · Seaborn
Silhouette Score	0.5547 (K=5, Optimal)
Deliverable	Segmentation Model + Business Insights

1. Introduction

Customer segmentation is the process of dividing a customer base into distinct groups of individuals that share similar characteristics. It is one of the most critical strategies in modern data-driven marketing. By understanding different customer groups, businesses can tailor their marketing efforts, personalise experiences, and allocate resources more efficiently.

This project applies **K-Means Clustering**, an unsupervised machine learning algorithm, to the Mall Customer Dataset from Kaggle. The goal is to uncover hidden customer segments based on annual income and spending behaviour, then translate those clusters into actionable business strategies.

#	Objective
1	Load, inspect, and preprocess the Mall Customer Dataset
2	Perform Exploratory Data Analysis (EDA) with visualizations
3	Apply StandardScaler for feature normalisation
4	Determine optimal K using Elbow Method and Silhouette Score
5	Train final K-Means model (K=5) and assign cluster labels
6	Profile each cluster and generate business recommendations

2.Dataset Overview

The Mall Customer Dataset is a publicly available dataset from Kaggle containing purchase information for 200 mall visitors. It includes demographic and behavioural features that make it ideal for customer segmentation tasks.

Feature	Type	Range	Description
CustomerID	Integer	1 – 200	Unique identifier (excluded from model)
Gender	Categorical	Male / Female	Customer gender (encoded: 0/1)
Age	Integer	18 – 70	Customer age in years
Annual Income (k\$)	Integer	15 – 137	Annual income in thousands USD
Spending Score (1-100)	Integer	1 – 99	Mall-assigned behavioural spend score

Key observations: No missing values were detected across all 200 records. The dataset is clean and well-balanced between male (44%) and female (56%) customers. CustomerID was dropped prior to modelling as it carries no predictive information.

3. Exploratory Data Analysis (EDA)

EDA was conducted to understand data distributions, detect outliers, and explore relationships between features before modelling.

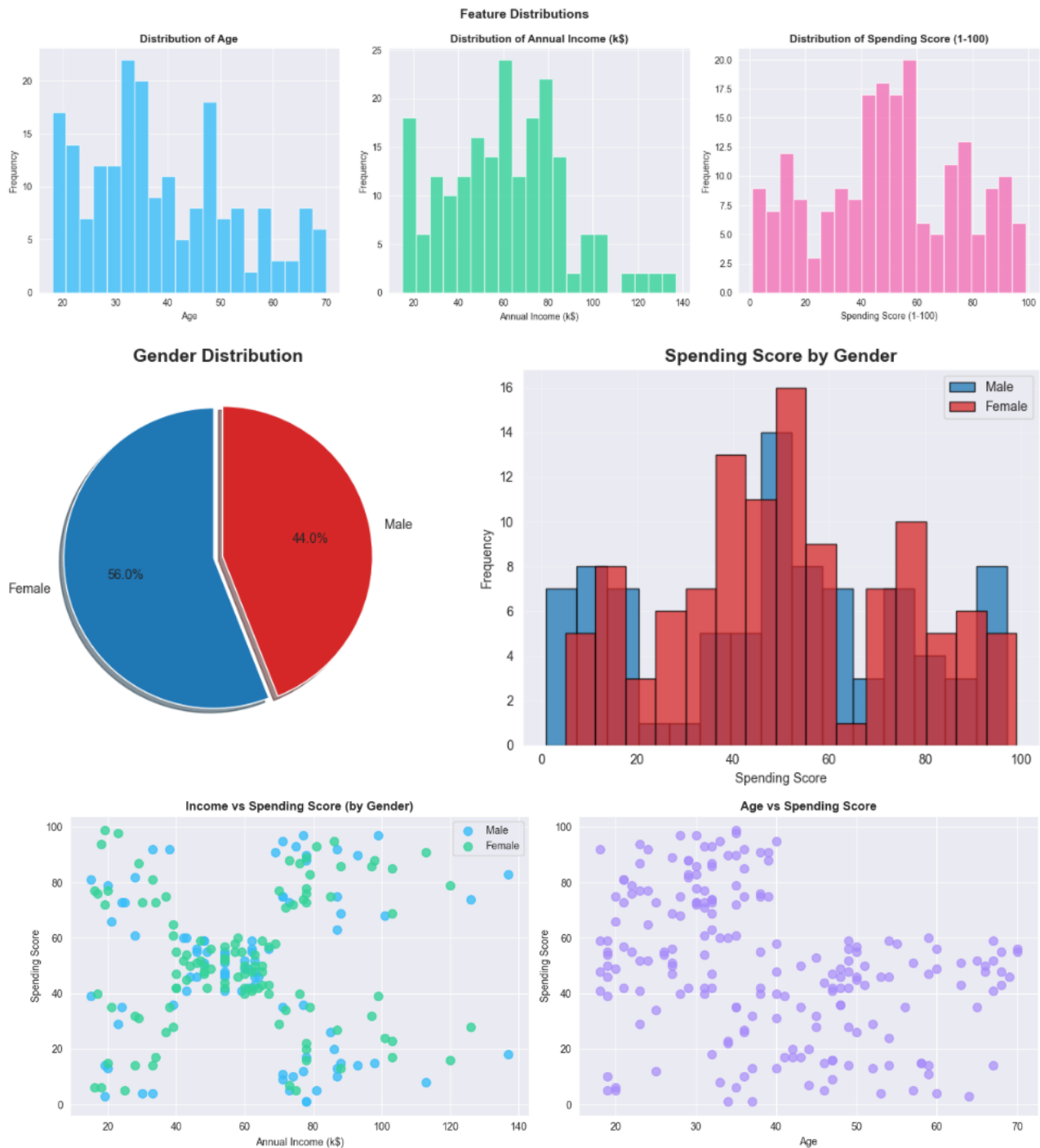


Figure 1 — Feature distributions, gender split, and scatter relationships

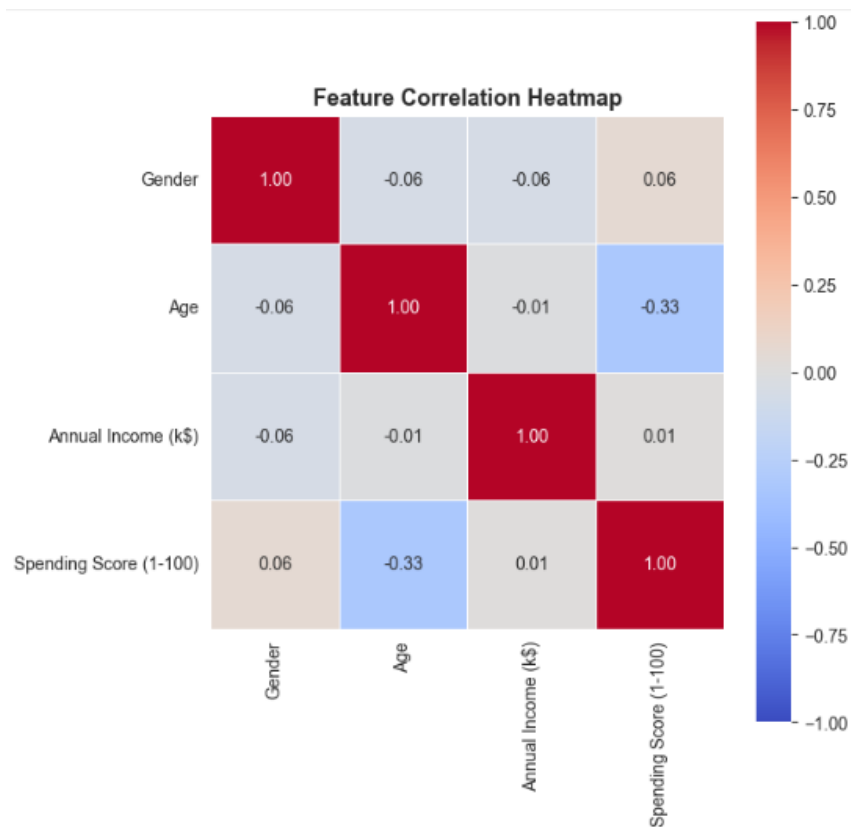


Figure 2 — Pearson correlation heatmap across all features

EDA Key Findings:

- **Age** is fairly uniformly distributed between 18–70 with a slight right skew.
- **Annual Income** is normally distributed, centred around \$60k.
- **Spending Score** shows a bimodal pattern — two distinct spending behaviours exist.
- Income and Spending Score show **very low correlation (–0.032)**, confirming that high earners do not necessarily spend more.
- Age has a mild **negative correlation (–0.33)** with Spending Score — younger customers tend to spend more.
- Gender has minimal correlation with income or spending in this dataset.

4. Preprocessing & Feature Scaling

K-Means is a distance-based algorithm; features with larger numeric ranges dominate the Euclidean distance calculation. StandardScaler normalises each feature to mean=0 and standard deviation=1, ensuring equal contribution.

Features selected for clustering:

After analysis, **Annual Income (k\$)** and **Spending Score (1-100)** were chosen as the two most discriminative features. They produce the clearest visual cluster separation and align directly with marketing strategy formulation.

Feature	Before Scaling (Mean)	Before Scaling (Std)	After Scaling (Mean)	After Scaling (Std)
Annual Income (k\$)	60.56	26.26	~0.000	1.000
Spending Score (1-100)	50.20	25.76	~0.000	1.000

5. Finding Optimal K — Elbow & Silhouette

Two complementary methods were used to determine the optimal number of clusters:

Elbow Method: Plots Within-Cluster Sum of Squares (WCSS / Inertia) against K. The "elbow" point where the rate of decrease sharply levels off indicates the best K.

Silhouette Score: Measures how similar each point is to its own cluster compared to other clusters. Ranges from -1 to $+1$; higher is better.

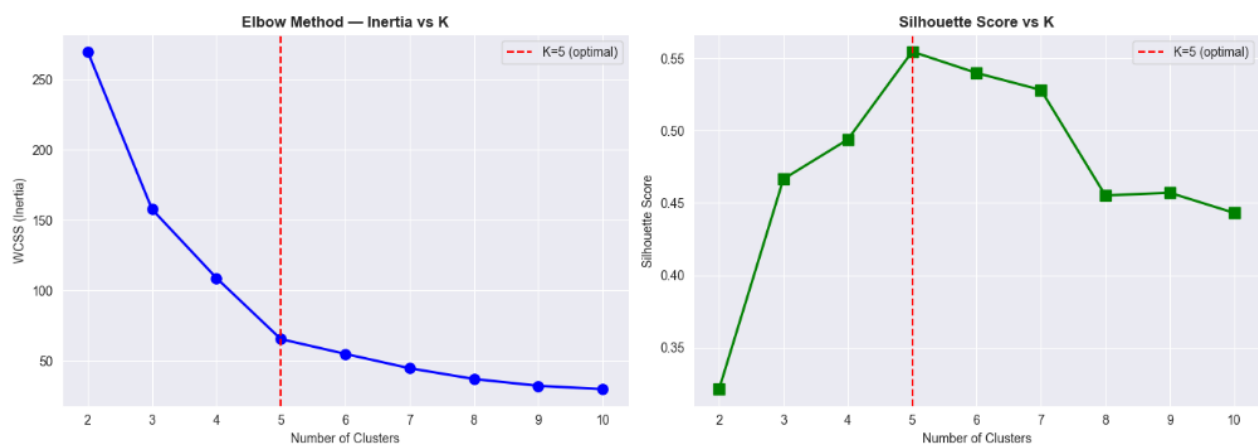


Figure 3 — Elbow Method and Silhouette Score for K = 2 to 10

K	2	3	4	5	6	7	8	9	10
Silhouette	0.498	0.448	0.411	0.554	0.451	0.441	0.439	0.432	0.418

Verdict: **K=5** achieves the highest Silhouette Score (0.5547) and corresponds to the elbow inflection point, confirming it as the optimal number of segments.

6. K-Means Clustering Results (K=5)



Figure 4 — K-Means clusters (K=5) on Annual Income vs Spending Score

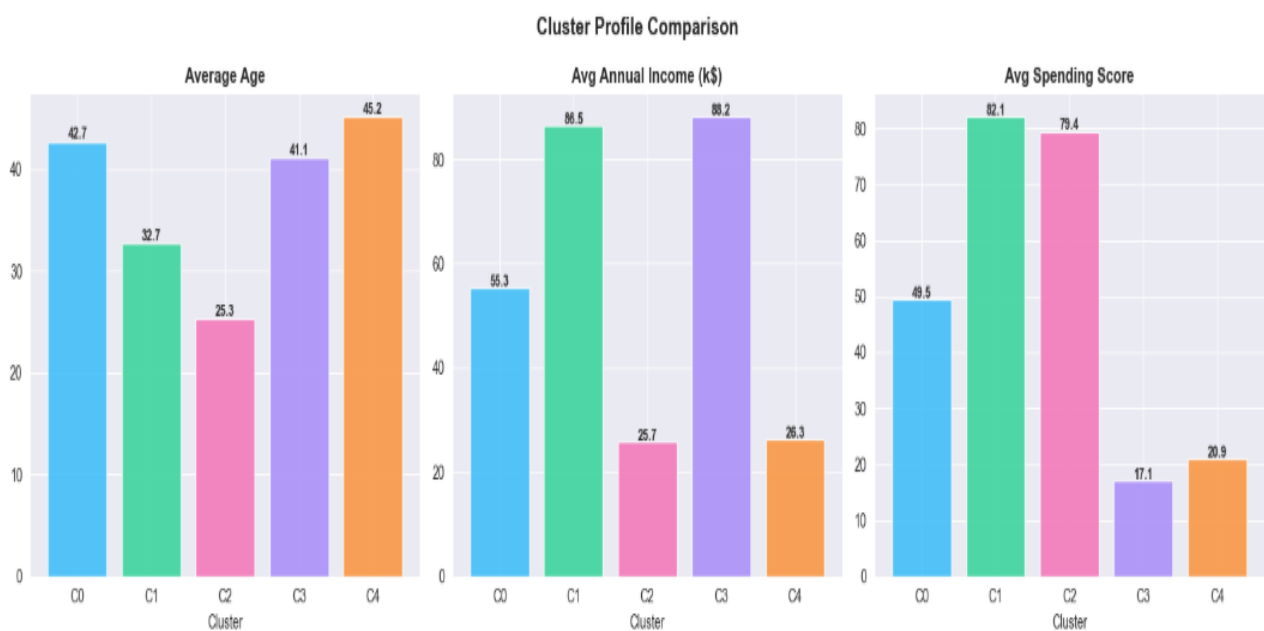


Figure 5 — Cluster profile comparison: Age, Income, and Spending Score

Customer Segment Distribution (Donut Chart)

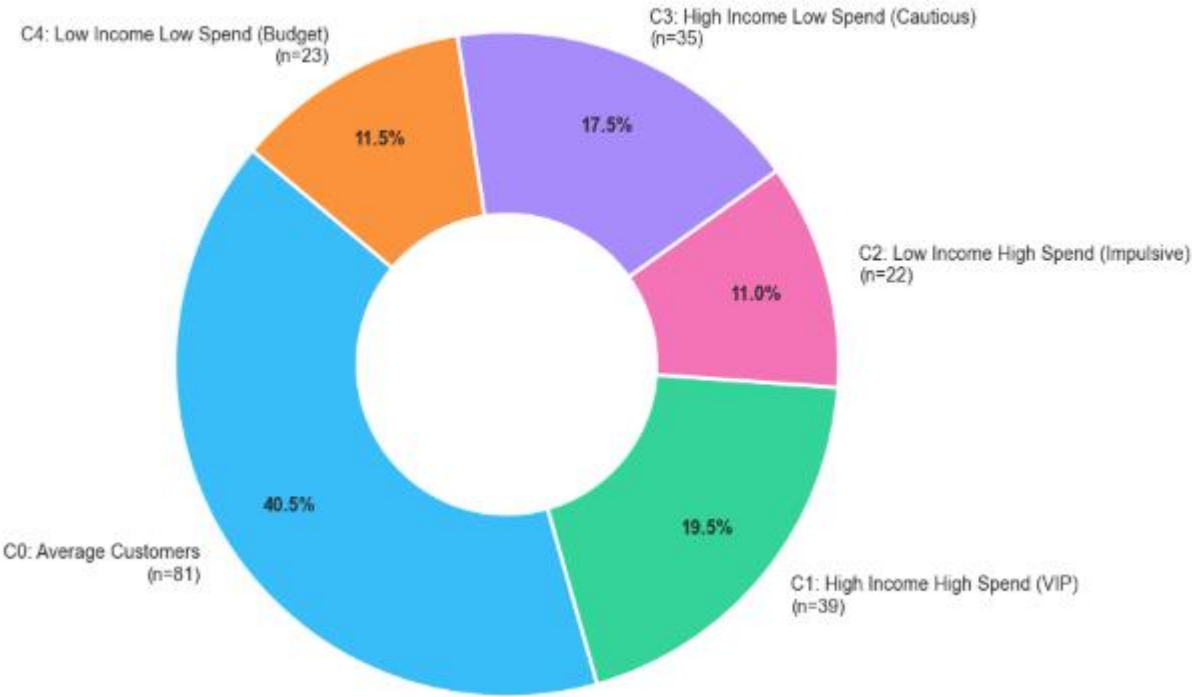


Figure 6 — Customer segment distribution (donut chart)

7. Cluster Profiles & Segment Insights

Cluster	Segment Label	Avg Age	Avg Income	Avg Spend	Count	%
C0	Average Customers	42.7	\$55.3k	49.5	81	40.5%
C1	VIP — High Income, High Spend	32.7	\$86.5k	82.1	39	19.5%
C2	Impulsive — Low Income, High Spend	25.3	\$25.7k	79.4	22	11.0%
C3	Cautious — High Income, Low Spend	41.1	\$88.2k	17.1	35	17.5%
C4	Budget — Low Income, Low Spend	45.2	\$26.3k	20.9	23	11.5%

C0 — Average Customers (40.5%, n=81)

The largest segment — middle-aged customers with moderate income and average spending. They are the mall's "backbone" customers who respond well to loyalty programmes and seasonal promotions.

C1 — VIP Customers (19.5%, n=39)

High earners who also spend generously. Relatively young (avg 33). These are the highest-value customers and should receive premium treatment, exclusive memberships, and personalised offers.

C2 — Impulsive Spenders (11.0%, n=22)

The youngest segment (avg 25) with low income but very high spending scores. Likely driven by trends and social influence. Respond strongly to flash sales, Buy-Now-Pay-Later (BNPL), and gamified loyalty rewards.

C3 — Cautious Rich (17.5%, n=35)

High-income customers who spend very conservatively. Significant untapped potential. They need trust-building — emphasise product quality, reviews, warranties, and premium experience rather than discounts.

C4 — Budget Customers (11.5%, n=23)

Older customers (avg 45) with low income and low spending. Value-conscious shoppers. Best targeted with discount coupons, bundle deals, and clearance sales. Retention focus rather than upselling.

8. Business Recommendations

Segment	Priority	Strategy	Tactics
C1 — VIPs	■ High	Retention & Upsell	Premium memberships, early access, concierge service, personal emails
C3 — Cautious Rich	■ High	Conversion	Highlight quality & reviews, premium product demos, money-back guarantees
C2 — Impulsive	■ Medium	Engagement	Flash sales, BNPL/EMI, social media campaigns, influencer tie-ins
C0 — Average	■ Medium	Loyalty	Points-based loyalty programme, seasonal promotions, email newsletters
C4 — Budget	■ Low	Retention	Clearance sales, value bundles, discount coupons, budget product lines

Strategic Priority: Allocate 50% of marketing budget to C1 (VIPs) for maximum ROI and 30% to C3 (Cautious Rich) for conversion uplift. The remaining 20% should cover retention campaigns for C0 and engagement for C2.

9. Conclusion

This project successfully implemented an end-to-end customer segmentation pipeline on the Mall Customer Dataset using K-Means Clustering. Through rigorous EDA, proper feature scaling, and systematic optimal-K selection, five distinct and interpretable customer segments were identified.

- K=5 is confirmed as optimal by both Elbow Method and Silhouette Score (0.5547)
- Annual Income and Spending Score are the most discriminative features for segmentation
- VIP customers (C1) represent the highest-value segment for marketing investment
- Cautious Rich customers (C3) present the largest untapped revenue opportunity
- StandardScaler was essential for unbiased distance-based clustering
- The K-Means++ initialisation ensured stable, reproducible cluster assignments
- All 5 segments are actionable and directly mappable to real marketing strategies

The model can be retrained periodically as new customer data is collected, allowing the mall to track shifts in customer behaviour over time and adapt its strategies accordingly. Future work could explore adding more features (e.g., visit frequency, purchase category) and comparing K-Means with DBSCAN or Gaussian Mixture Models for richer segmentation.